ISO-Style Process Sheet

LEA-Hyper-Alloy – Manufacturing Process Specification

1 Scope

This process sheet specifies the manufacturing steps, temperature and time windows, and quality requirements for the production of a conductive aluminum-copper-magnesium alloy (LEA alloy).

The process applies to continuously cast, direct-chill cast, extruded, and rolled semi-finished products.

2 Normative References

The following referenced documents are indispensable for the application of this process sheet:

- **EN 573** – Aluminium and aluminium alloys – Chemical composition
- **EN 485** – Aluminium and aluminium alloy sheet, strip and plate – Mechanical properties
- **ISO 6892-1** – Metallic materials – Tensile testing
- **ISO 6506-1** – Metallic materials – Brinell hardness test
- **ISO 1463** – Metallic coatings – Micrographic determination of thickness

3 Terms and Definitions

LEA alloy:

An aluminum-based alloy containing 1.5–3.5 wt% Cu and 1.0–2.0 wt% Mg, designed for enhanced electrical conductivity combined with moderate to high mechanical strength.

4 Chemical Composition

The alloy shall comply with the following mass fraction limits:

Element	Range (wt%)
Aluminium (Al)	93.5–96.5
Copper (Cu)	1.5–3.5
Magnesium (Mg)	1.0–2.0
Trace elements (Fe, Si, Mn, etc.)	≤ 0.2

5 Process Flow

5.1 Overview (formal arrow diagram)

Raw Material Preparation
↓
Melting (650–700 °C, 30–120 min)
↓
Casting (680–700 °C)
↓
Homogenization (optional: 480–520 °C, 4–12 h)
↓
Hot Forming (400–450 °C)
↓
Solution Heat Treatment (500–520 °C, 30–90 min)
↓
Quenching (water or air)
↓
Artificial Aging / Overaging (150–200 °C, 4–12 h)
↓
Final Machining
↓
Quality Inspection

5.2 Detailed Process Requirements

5.2.1 Melting

- Temperature: 650–700 °C
- Holding time: 30–120 min
- Atmosphere: air or optional inert gas (argon)
- Objective: complete dissolution of alloying elements

5.2.2 Casting

- Casting temperature: 680–700 °C
- Methods: continuous casting or direct-chill casting
- Cooling: controlled, air- or mold-cooled

5.2.3 Homogenization (optional but recommended)

- Temperature: 480–520 °C
- Duration: 4–12 h
- Purpose: reduction of segregation and microstructural homogenization

5.2.4 Hot Forming

- Entry temperature: 380–450 °C
- Process temperature: 400–450 °C
- Methods: extrusion or hot rolling

5.2.5 Solution Heat Treatment

- Temperature: 500–520 °C
- Duration: 30–90 min
- Purpose: formation of a supersaturated solid solution

5.2.6 Quenching

- Medium: water (10–40 °C) or air
- Purpose: retention of supersaturation

5.2.7 Artificial Aging / Overaging

- Temperature: 150–200 °C
- Duration: 4–12 h
- Purpose: adjustment of conductivity/strength balance
- Recommended settings:
- LEA-90: higher T, longer t → maximum conductivity
- LEA-100: intermediate T/t → balanced properties
- LEA-120: lower T, shorter t → higher strength

6 Quality Requirements

6.1 Mechanical Properties

- Tensile strength (Rm): 180–320 MPa (depending on grade)
- Elongation (A): ≥ 8 %
- Hardness: 60–90 HB

6.2 Electrical Properties

- Electrical conductivity: 45–55 % IACS (depending on grade)

6.3 Microstructural Requirements

- Uniform grain structure
- Low porosity
- No coarse intermetallic precipitates
- Controlled precipitation morphology

7 Inspection and Testing

- Tensile test: ISO 6892-1
- Hardness test: ISO 6506-1
- Electrical conductivity: eddy-current method
- Metallography: optical or SEM examination
- Dimensional inspection: per product specification

8 Marking and Traceability

Each batch shall be marked with:

- Alloy grade (LEA-90 / LEA-100 / LEA-120)
- Melt number
- Production date
- Heat treatment condition
- Verified test results (Rm, A, HB, IACS)